

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 2)

If frequency of electromagnetic wave is 60MHz and it travels in air along z direction then the corresponding electric and magnetic field vectors will be mutually perpendicular to each other and the wavelength of the wave (in m) is:

(1) 2.5

(2) 10

(3) 5

(4) 2

Q2 - 2024 (27 Jan Shift 1)

A plane electromagnetic wave propagating in x-direction is described by

$$E_y = (200 \text{Vm}^{-1}) \sin [1.5 \times 10^7 t - 0.05x];$$

The intensity of the wave is :

(Use $\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$)

(1) 35.4Wm^{-2}

(2) 53.1Wm^{-2}

(3) 26.6Wm^{-2}

(4) 106.2Wm^{-2}

Q3 - 2024 (27 Jan Shift 2)

An object is placed in a medium of refractive index 3. An electromagnetic wave of intensity $6 \times 10^8 \text{W/m}^2$ falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space $= 3 \times 10^8 \text{m/s}$):

(1) 36Nm^{-2}

(2) 18Nm^{-2}

(3) 6Nm^{-2}

(4) 2Nm^{-2}

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Q4 - 2024 (29 Jan Shift 2)

A plane electromagnetic wave of frequency 35 MHz travels in free space along the X-direction. At a particular point (in space and time) $\vec{E} = 9.6\hat{j}\text{V/m}$. The value of magnetic field at this point is :

(1) $3.2 \times 10^{-8}\hat{k}\text{T}$

(2) $3.2 \times 10^{-8}\hat{i}\text{T}$

(3) $9.6\hat{j}\text{T}$

(4) $9.6 \times 10^{-8}\hat{k}\text{T}$

Q5 - 2024 (30 Jan Shift 1)

The electric field of an electromagnetic wave in free space is represented as $\vec{E} = E_0 \cos(\omega t - kz)\hat{i}$.

The corresponding magnetic induction vector will be :

(1) $\vec{B} = E_0 C \cos(\omega t - kz)\hat{j}$

(2) $\vec{B} = \frac{E_0}{C} \cos(\omega t - kz)\hat{j}$

(3) $\vec{B} = E_0 \cos(\omega t + kz)\hat{j}$

(4) $\vec{B} = \frac{E_0}{C} \cos(\omega t + kz)\hat{j}$

Q6 - 2024 (30 Jan Shift 2)

Match List I with List II

List-I		List-II	
A.	Gauss's law of magnetostatics	I.	$\oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho dV$
B.	Faraday's law of electro magnetic induction	II.	$\oint \vec{B} \cdot d\vec{a} = -0$
C.	Ampere's law	III.	$\oint \vec{E} \cdot d\vec{l} = \frac{-d}{dt} \int \vec{B} \cdot d\vec{a}$
D.	Gauss's law of electrostatics	IV.	$\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$

Choose the correct answer from the options given below:

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(1) A-I, B-III, C-IV, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-II, C-III, D-I

(4) A-II, B-III, C-IV, D-I

Q7 - 2024 (31 Jan Shift 1)

In a plane EM wave, the electric field oscillates sinusoidally at a frequency of 5×10^{10} Hz and an amplitude of 50 Vm^{-1} . The total average energy density of the electromagnetic field of the wave is : [Use

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2]$$

(1) $1.106 \times 10^{-8} \text{ Jm}^{-3}$ (2) $4.425 \times 10^{-8} \text{ Jm}^{-3}$ (3) $2.212 \times 10^{-8} \text{ Jm}^{-3}$ (4) $2.212 \times 10^{-10} \text{ Jm}^{-3}$

Q8 - 2024 (31 Jan Shift 2)

Given below are two statements:

Statement I: Electromagnetic waves carry energy as they travel through space and this energy is equally shared by the electric and magnetic fields.

Statement II: When electromagnetic waves strike a surface, a pressure is exerted on the surface.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is incorrect but Statement II is correct.

(2) Both Statement I and Statement II are correct.

(3) Both Statement I and Statement II are incorrect.

(4) Statement I is correct but Statement II is incorrect.

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Questions with Answer Keys

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Answer Key

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Q1 (3) mathongo /// ma**Q2** (2) mathongo **Q3** (3) mathongo /// mc**Q4** (1) mathongo

Q5 (2) mathongo /// ma**Q6** (4) mathongo **Q7** (1) mathongo /// mc**Q8** (2) mathongo

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Solutions

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Q1

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{60 \times 10^6} = 5 \text{ m}$$

Q2

$$I = \frac{1}{2} \epsilon_0 E_0^2 \times c$$

$$I = \frac{1}{2} \times 8.85 \times 10^{-12} \times 4 \times 10^4 \times 3 \times 10^8$$

$$I = 53.1 \text{ W/m}^2$$

Q3

$$\text{Radiation pressure} = \frac{I}{v}$$

$$= \frac{I \cdot \mu}{c}$$

$$= \frac{6 \times 10^8 \times 3}{3 \times 10^8}$$

$$= 6 \text{ N/m}^2$$

Q4

$$\frac{E}{B} = C$$

$$\frac{E}{B} = 3 \times 10^8$$

$$B = \frac{E}{3 \times 10^8} = \frac{9.6}{3 \times 10^8}$$

$$B = 3.2 \times 10^{-8} \text{ T}$$

$$\hat{B} = \hat{v} \times \hat{E}$$

$$= \hat{i} \times \hat{j} = \hat{k}$$

So,

$$\vec{B} = 3.2 \times 10^{-8} \hat{k} \text{ T}$$

Q5

$$\text{Given } \vec{E} = E_0 \cos(\omega t - kz) \hat{i}$$

$$\vec{B} = \frac{E_0}{C} \cos(\omega t - kz) \hat{j}$$

$$\hat{C} = \hat{E} \times \hat{B}$$

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Solutions

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Q6

Maxwell's equation

Q7

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

$$U_E = \frac{1}{2} \times 8.85 \times 10^{-12} \times (50)^2$$
$$= 1.106 \times 10^{-8} \text{ J/m}^3$$

Q8

$$\frac{1}{2} \epsilon_0 E^2 = \frac{B^2}{2\mu_0}$$

$$\therefore E = CB \text{ and } C = \frac{1}{\mu_0 \epsilon_0}$$

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